

Dylan Dsouza

San Diego, California

✉ dydsouza@ucsd.edu

🌐 dylandsouza.com

🌐 linkedin.com/in/dsouza-dylan

🐙 github.com/dsouza-dylan

EDUCATION

Bachelor of Science, Data Science | University of California San Diego | GPA: 3.94 – Provost Honors Expected Mar 2027
Relevant Coursework: Applied Data Science, Machine Learning Theory, Recommender Systems, Statistical Analysis

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C/C++, HTML/CSS, JavaScript

Libraries/Frameworks: NumPy, pandas, scikit-learn, PyTorch, TensorFlow, OpenCV, Plotly, FinBERT, Jupyter, Git/GitHub

Core Competencies: Data Cleaning/Visualization, EDA, Web Scraping, A/B Testing, Financial Modeling, Time Series Analysis

EXPERIENCE

Quantitative Analyst Oct 2025 – Present
Sustainable Investment Group at UC San Diego San Diego, CA

- Developed Python web scraping framework using FinBERT NLP, Yahoo Finance, and Tavily to generate investment signals by analyzing lagged relationships between 25 weeks of news sentiment and 5+ market metrics (returns, volatility, volume).
- Co-authored an Under Armour investment thesis for a 5-team stock pitch competition, combining ARIMA and sentiment analysis models with macroeconomic analysis, ESG factors, and DCF valuation to deliver a data-driven sell recommendation.

Machine Learning Researcher Oct 2025 – Present
Engineers for Exploration San Diego, CA

- Developing ensemble machine learning models (70% mIoU baseline) for mangrove mapping, including image segmentation pipelines and preprocessing of satellite and drone imagery for pixel-level land-cover classification.
- Curating 7 public datasets (SpaceNet, xView, LandCover.ai, etc.) to guide early-stage model design, inform feature selection, and aid ML-assisted labeling workflows for 4 classes (road, building, water, and mangrove).

Undergraduate Researcher – Marine Data Science May 2025 – Present
Scripps Institution of Oceanography (Dr. Jennifer Smith Lab) San Diego, CA

- Generated scaled and oriented photo quadrats using Viscore and labeled 9,000+ georeferenced points of coral imagery to produce a structured dataset for downstream spatial and temporal analysis of 15 reef sites on Palmyra Atoll.
- Conducting exploratory and time series analyses across 16 time points (2014–2025) to investigate depth and treatment effects on reef dynamics, identifying temporal trends and coral bleaching extent to inform restoration strategies.

Board Member – Online Content Oct 2024 – Present
Data Science Student Society at UC San Diego San Diego, CA

- Research and author data-driven articles on machine learning applications, translating complex technical topics into accessible narratives for a student audience comprising 2,350+ followers and 500+ community members.
- Featured how predictive modeling, photomosaic imaging, and machine learning advance coral reef restoration and wildfire detection, highlighting projects from NASA, UC Berkeley, and UC San Diego's WIFIRE Lab.

Marketing Intern Jul 2023 – Aug 2023
Bennett Coleman & Co. Ltd. (The Times of India) Mumbai, India

- Conducted quantitative research on 56,000+ newspaper headlines using TF-IDF vectorization and multi-output linear regression, training a model to predict relevance scores across 5 key market segments.
- Delivered 5-category classification prototype achieving 85% accuracy, enabling automated headline categorization across Business, Politics, Technology, Social Issues, and Lifestyle segments for editorial workflows.

PROJECTS

ENSOCast: Interactive Climate Dashboard May 2025 – Oct 2025
Personal Project (Presented at San Diego Undergraduate Tech Conference 2025) [View Project](#)

- Built a Streamlit dashboard with interactive Plotly visualizations for exploring 40+ years of ocean temperature data and ENSO phases, enabling users to train and compare custom ML models.
- Developed and evaluated advanced ML models, including Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble Learning, using temporal, seasonal, and interaction-based features, achieving up to 82% accuracy in ENSO phase prediction.

Power Outage Severity Prediction Aug 2025 – Sep 2025
Personal Project [View Project](#)

- Constructed an end-to-end machine learning pipeline on U.S. power outage data (2000–2016), including data cleaning, feature engineering, exploratory data analysis, and hypothesis/permutation testing on 1,500+ outage events.
- Designed and optimized a Random Forest model (after baseline Decision Tree), achieving 80% accuracy with balanced precision/recall for outage duration prediction.